Improved zero-density estimates for the Riemann zeta function

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We prove improved zero-density estimates for the Riemann zeta function, strengthening our previously announced bounds in several ranges. Let N(σ,T) count the nontrivial zeros ρ with Re ρ ≥ σ and |Im ρ| ≤ T, counted with multiplicity. For T ≥ 2 and ε > 0, we obtain bounds of the form N(σ,T) ≤ C(σ,ε)T^((1−σ)A(σ)+ε). In particular, we may take A(σ)=12/(9σ−1) for 5/7 ≤ σ ≤ 113/153 and A(σ)=255/(153σ+7) for 113/153 ≤ σ ≤ 3/4. We also obtain a sequence of bounds for 7/9 < σ ≤ 7/8. The complete estimates and their ranges are given in an accompanying table and compared graphically with earlier bounds. The results are formalized in Lean.

$$N(\sigma, T) \leq C_{\sigma,\varepsilon}\, T^{p(\sigma)+\varepsilon}, \qquad p(\sigma) = A(\sigma)(1-\sigma)$$

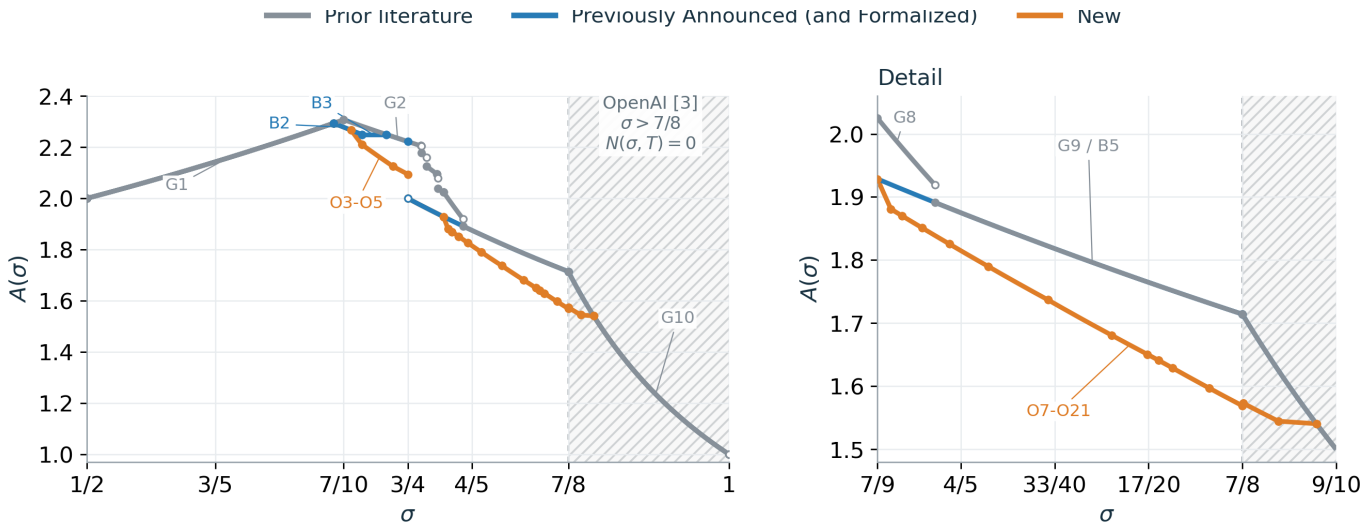


Figure 1. The three curves are cumulative minima of the listed bounds: Prior literature (gray), Previously Announced (and Formalized) (blue), and New (orange). Coincident portions use the earlier color. Exact formulas and endpoints are listed on pages 4-5.

Neutral hatching cites OpenAI [3]: N(σ,T)=0 for σ > 7/8; the boundary is excluded. This checked input is credited to OpenAI. Segment IDs and exact formulas are on pages 4-5.

The plotted power curves compare the named earlier submissions; “New” denotes additions to that comparison. Hatching marks the imported OpenAI input at σ>7/8. Later zero-free refinements are cited in the text.

The 73 main density statements passed protected Comparator with Lean, NanoDa and verified con-ron. Page 6 reports separate 4.34.1 supplements. References: [1], [2], [3].

Table 1. Proved contribution bounds

Write p(σ) = (1 - σ)A(σ), except where p is given directly. S means C may depend on fixed σ and ε. U means one Cε works for every σ in the printed range. In each case C is chosen before all real T ≥ 2.

ID	A(σ), or p(σ)	Full σ range	C	Attribution
P11	$A = \frac{18}{7\sigma+3}$	$\left[\frac{9}{13}, \frac{5}{7}\right]$	S	Lewko announced
P12	$A = \frac{9}{4}$	$\left[\frac{5}{7}, \frac{3}{4}\right]$	S	Lewko announced
P13	$A = \frac{3}{2\sigma}$	$\left(\frac{3}{4}, 1\right)$	S	Lewko announced
P14	$A = \frac{6}{8\sigma-3}$	$\left[\frac{7}{10}, \frac{5}{7}\right]$	S	Lewko † domination / native
P15	$A = \frac{12}{9\sigma-1}$	$\left[\frac{5}{7}, \frac{113}{153}\right]$	S	Lewko new
P16	$A = \frac{255}{153\sigma+7}$	$\left[\frac{113}{153}, \frac{3}{4}\right]$	S	Lewko new
P17	$A = \frac{3}{11\sigma-7}$	$\left[\frac{7}{9}, \frac{29359}{37574}\right]$	S	Lewko new
P18	$A = \frac{4744}{4870\sigma-1283}$	$\left[\frac{29359}{37574}, \frac{1015}{1294}\right]$	S	Lewko new
P19	$A = \frac{6154}{6354\sigma-1693}$	$\left[\frac{1015}{1294}, \frac{353}{447}\right]$	S	Lewko new
P20	$A = \frac{3782}{3936\sigma-1065}$	$\left[\frac{353}{447}, \frac{467}{586}\right]$	S	Lewko new
P21	$A = \frac{2168}{2282\sigma-631}$	$\left[\frac{467}{586}, \frac{289}{358}\right]$	S	Lewko new
P22	$A = \frac{566}{606\sigma-173}$	$\left[\frac{289}{358}, \frac{163}{198}\right]$	S	Lewko new
P23	$A = \frac{31544}{35670\sigma-11203}$	$\left[\frac{163}{198}, \frac{7873}{9370}\right]$	S	Lewko new
P24	$A = \frac{1261710}{1447460\sigma-465509}$	$\left[\frac{7873}{9370}, \frac{2018}{2375}\right]$	S	Lewko new
P25	$A = \frac{3066}{3616\sigma-1215}$	$\left[\frac{2018}{2375}, \frac{1959705}{2298302}\right]$	S	Lewko new
P26	$A = \frac{597962}{729216\sigma-257415}$	$\left[\frac{1959705}{2298302}, \frac{6482825}{7570038}\right]$	S	Lewko new
P27	$A = \frac{16504356}{20732766\sigma-7623595}$	$\left[\frac{6482825}{7570038}, \frac{2905}{3354}\right]$	S	Lewko new
P28	$A = \frac{2172}{2742\sigma-1015}$	$\left[\frac{2905}{3354}, \frac{7}{8}\right]$	S	Lewko new
P29	$p = \frac{1086}{5537}$	$\left[\frac{7}{8}, 1\right]$	U	Lewko monotonicity
P30	$A = \frac{24}{30\sigma-11}$	$\left[\frac{19}{22}, \frac{23}{26}\right]$	U	Lewko † native / zero-free
P31	$A = \frac{6}{\sigma+3}$	$\left[\frac{23}{26}, 1\right]$	U	Lewko † zero-free consequence

Also proved: the previously announced global bound

A(σ) = 39/17 on the closed interval [1/2,1], with a uniform Cε. This weaker bound is retained here and is inactive in the cumulative frontier.

† Existing-bound or zero-free consequences; see page 3. All printed bounds and ranges remain valid.

Historical comparisons and constants

These ten rows are comparisons with prior literature. The full proved source ranges are shown, even where the corresponding curve is not the minimum.

ID	A(σ), or p(σ)	Full σ range	C	Attribution
P1	$A = \frac{3}{2-\sigma}$	$\left[\frac{1}{2}, \frac{7}{10}\right]$	S	Ingham
P2	$A = \frac{15}{5\sigma+3}$	$\left[\frac{7}{10}, \frac{19}{25}\right)$	S	Guth-Maynard
P3	$A = \frac{9}{8\sigma-2}$	$\left[\frac{19}{25}, \frac{127}{167}\right)$	S	Ivić / TTY Table 2
P4	$A = \frac{15}{13\sigma-3}$	$\left[\frac{127}{167}, \frac{13}{17}\right)$	S	TTY Table 2 row 4
P5	$A = \frac{6}{5\sigma-1}$	$\left[\frac{13}{17}, \frac{17}{22}\right)$	S	TTY Table 2 row 5
P6	$A = \frac{2}{9\sigma-6}$	$\left[\frac{17}{22}, \frac{41}{53}\right)$	S	TTY Table 2 row 6
P7	$A = \frac{9}{7\sigma-1}$	$\left[\frac{41}{53}, \frac{7}{9}\right)$	S	TTY Table 2 row 7
P8	$A = \frac{9}{16\sigma-8}$	$\left[\frac{7}{9}, \frac{1867}{2347}\right)$	S	TTY Table 2 row 8
P9	$A = \frac{3}{2\sigma}$	$\left[\frac{23}{29}, 1\right]$	S	Kerr Thm. 6
P10	$A = \frac{3}{10\sigma-7}$	$\left(\frac{7}{10}, 1\right]$	S	Heath-Brown

Constant dependence

S: for each fixed σ in the range and each ε > 0, there is C(σ,ε) > 0 such that the bound holds for every real T ≥ 2.

U: for each ε > 0, there is Cε > 0 such that the bound holds simultaneously for every σ in the printed range and every real T ≥ 2. This is proved for P29-P31 and the global 39/17 bound.

The announced 9/4 row is stated with C(σ,ε); no uniform constant is asserted for that row.

Square brackets include the endpoint; round brackets exclude it. At σ = 1 the actual zero count is zero. The plotted normalized A is a left limit.

Attribution and proof routes

Historical comparisons use stronger native bounds and OpenAI [3] where applicable. The curve A=6/(8σ-3) uses announced-bound domination on [7/10,12/17] and the native two-term bound on (12/17,5/7]. The curve A=24/(30σ-11) uses native domination through 7/8 and zero-freeness above it. A=6/(σ+3) on [23/26,1] is a zero-free consequence. Historical rows A=3/(2σ) and A=3/(10σ-7) also use zero-freeness above 7/8. These are checked conclusions; the original historical arguments are not claimed here.

Exact frontier: prior and announced

The IDs agree with the figure. These are active intervals, not the full source ranges on pages 2-3. An endpoint may be supplied by an equal-valued adjacent theorem. At σ = 1 only the left limit of A is plotted.

IDs	A(σ)	Active σ interval
G1 P1	$\frac{3}{2-\sigma}$	$[\frac{1}{2}, \frac{7}{10}]$
G2 P2	$\frac{15}{5\sigma+3}$	$[\frac{7}{10}, \frac{19}{25}]$
G3 P3	$\frac{9}{8\sigma-2}$	$[\frac{19}{25}, \frac{127}{167})$
G4 P4	$\frac{15}{13\sigma-3}$	$[\frac{127}{167}, \frac{13}{17})$
G5 P5	$\frac{6}{5\sigma-1}$	$[\frac{13}{17}, \frac{17}{22}]$
G6 P6	$\frac{2}{9\sigma-6}$	$[\frac{17}{22}, \frac{41}{53})$
G7 P7	$\frac{9}{7\sigma-1}$	$[\frac{41}{53}, \frac{7}{9}]$
G8 P8	$\frac{9}{16\sigma-8}$	$[\frac{7}{9}, \frac{23}{29})$
G9 P9	$\frac{3}{2\sigma}$	$[\frac{23}{29}, \frac{7}{8}]$
G10 P10	$\frac{3}{10\sigma-7}$	$[\frac{7}{8}, 1)$
B1 P1	$\frac{3}{2-\sigma}$	$[\frac{1}{2}, \frac{9}{13}]$
B2 P11	$\frac{18}{7\sigma+3}$	$[\frac{9}{13}, \frac{5}{7}]$
B3 P12	$\frac{9}{4}$	$[\frac{5}{7}, \frac{11}{15}]$
B4 P2	$\frac{15}{5\sigma+3}$	$[\frac{11}{15}, \frac{3}{4}]$
B5 P13	$\frac{3}{2\sigma}$	$(\frac{3}{4}, \frac{7}{8}]$
B6 P10	$\frac{3}{10\sigma-7}$	$[\frac{7}{8}, 1)$

Exact frontier: new cumulative bounds

The IDs agree with the figure. These are active intervals, not the full source ranges on pages 2-3. An endpoint may be supplied by an equal-valued adjacent theorem. At σ = 1 only the left limit of A is plotted.

IDs	A(σ)	Active σ interval
O1 P1	$\frac{3}{2-\sigma}$	$[\frac{1}{2}, \frac{9}{13}]$
O2 P11	$\frac{18}{7\sigma+3}$	$[\frac{9}{13}, \frac{12}{17}]$
O3 P14	$\frac{6}{8\sigma-3}$	$[\frac{12}{17}, \frac{5}{7}]$
O4 P15	$\frac{12}{9\sigma-1}$	$[\frac{5}{7}, \frac{113}{153}]$
O5 P16	$\frac{255}{153\sigma+7}$	$[\frac{113}{153}, \frac{3}{4}]$
O6 P13	$\frac{3}{2\sigma}$	$(\frac{3}{4}, \frac{7}{9}]$
O7 P17	$\frac{3}{11\sigma-7}$	$[\frac{7}{9}, \frac{29359}{37574}]$
O8 P18	$\frac{4744}{4870\sigma-1283}$	$[\frac{29359}{37574}, \frac{1015}{1294}]$
O9 P19	$\frac{6154}{6354\sigma-1693}$	$[\frac{1015}{1294}, \frac{353}{447}]$
O10 P20	$\frac{3782}{3936\sigma-1065}$	$[\frac{353}{447}, \frac{467}{586}]$
O11 P21	$\frac{2168}{2282\sigma-631}$	$[\frac{467}{586}, \frac{289}{358}]$
O12 P22	$\frac{566}{606\sigma-173}$	$[\frac{289}{358}, \frac{163}{198}]$
O13 P23	$\frac{31544}{35670\sigma-11203}$	$[\frac{163}{198}, \frac{7873}{9370}]$
O14 P24	$\frac{1261710}{1447460\sigma-465509}$	$[\frac{7873}{9370}, \frac{2018}{2375}]$
O15 P25	$\frac{3066}{3616\sigma-1215}$	$[\frac{2018}{2375}, \frac{1959705}{2298302}]$
O16 P26	$\frac{597962}{729216\sigma-257415}$	$[\frac{1959705}{2298302}, \frac{6482825}{7570038}]$
O17 P27	$\frac{16504356}{20732766\sigma-7623595}$	$[\frac{6482825}{7570038}, \frac{2905}{3354}]$
O18 P28	$\frac{2172}{2742\sigma-1015}$	$[\frac{2905}{3354}, \frac{7}{8}]$
O19 P29	$\frac{1086}{5537(1-\sigma)}$	$[\frac{7}{8}, \frac{24139}{27578}]$
O20 P30	$\frac{24}{30\sigma-11}$	$[\frac{24139}{27578}, \frac{23}{26}]$
O21 P31	$\frac{6}{\sigma+3}$	$[\frac{23}{26}, \frac{17}{19}]$
O22 P10	$\frac{3}{10\sigma-7}$	$[\frac{17}{19}, 1)$

Bounds for separate objects

These are separate Lean 4.34.1 supplements, outside the main 73-statement 4.35 submission. Their source and existing kernel receipts were rechecked, with fresh proof-closure inspection. Both zero subsets lie in the positive closed slab T ≤ Im ρ ≤ 2T. They are not full-density estimates for N(σ,T).

Polynomial large values at σ = 17/20

$$|W| \le C_\varepsilon T^{\frac{2817}{11744}+\varepsilon}$$

$$\sigma = \frac{17}{20}, \quad T^{\frac{1565}{2936}} \le N \le T^{\frac{4695}{5872}}$$

The polynomial length N lies in the displayed closed range. Coefficients are bounded by 1; translates are arbitrary and the sample points are one-separated, with polynomial modulus at least N to the power 17/20. For each ε > 0, Cε is chosen before all physical data and real T ≥ 2.

Positive canonical LONG zeros

$$R_{\mathrm{LONG}} \le C\, T^{\frac{395(1-\sigma)}{132(11\sigma-7)}+\varepsilon}$$

$$[L,U] \subset \left(\frac{2243}{2904}, \frac{3301}{4228}\right)$$

For every closed [L,U] strictly inside (2243/2904,3301/4228), choose δ(U) > 0 before η, ε, σ and T. For fixed L and U, C(L,U,η,ε) is then chosen before every σ in [L,U] and every real T ≥ 2. Analytic multiplicity is retained. In particular [7/9,1561/2000] is a closed admissible interval.

Positive canonical SHORT zeros at σ = 17/20

$$R_{\mathrm{SHORT}} \le C_{\delta,\varepsilon} T^{\frac{2817}{11744}+\varepsilon}$$

$$\sigma = \frac{17}{20}, \quad a = \frac{1565}{2936}$$

The detector exponent is a = 1565/2936. Choose η(ε) = min(1/16384, ε/2404) before each fixed positive smoothing δ; then choose C(δ,ε) before all real T ≥ 2. SHORT means membership in the original admissible normalized powered-polynomial family; LONG is its exact complement. Analytic multiplicity is retained.

Appendix: source and verification

Source scope

The 21 contribution rows, ten historical rows, weaker global bound, and three separate-object estimates are driven by the existing checked inventory. No proof source, mathematical range or exponent is changed by this presentation.

Imported input

OpenAI [3] supplies the strict zero-free half-plane Re s > 7/8. Its imported proof is kernel-checked here and credited to OpenAI. Upstream notices are retained.

Later literature

Subsequent Lean-verified refinements lower the zero-free boundary below 7/8, including Hailey Collet's strict bound Re(s)>874861/1000000=0.874861 [Collet26]. Liu's intermediate refinement gives B∞=0.874957019420098946... [Liu26]. Our comparison near 7/8 is with the specified earlier results and does not improve the current zero-free region.

Literature verification scope

The imported, kernel-checked input used by this formalization remains OpenAI's strict 7/8 theorem. Collet's result has separate public tracker verification with Lean, NanoDa and con-ron. Liu's repository reports a strict Lean source replay using pinned external artifacts. These later results are cited as literature; they are not imported into this release's 73-statement proof closure, and no new replay of them was run for this editorial revision.

Current checks

The 73 main statements on Lean 4.35.0-rc2 / Mathlib 065356127b1dc0016f66b7283ce0ce2c4055aa55 passed protected Comparator on 20 October 2026, including Lean, NanoDa and verified con-ron. Only Classical.choice, propext and Quot.sound are permitted. The proof sources and nine dependency pins are unchanged. Page 6 has separate 4.34.1 supplement coverage.

Verification scope

The protected Comparator result applies to the unchanged proof sources at snapshot 451872fb. Hosted Palomar verification, rendering and registry review have not run.

Open proposals

The mixed weighted trace estimate, the complementary short-detector saving near 39/50, the remaining LONG range at 17/20, and the density-hypothesis endpoint at 3/4 remain open. They are not plotted or counted as proved full-density results.

References

[1] PALOMAR-2026-10-11-000010 (v1), 11 October 2026. [2] PALOMAR-2026-10-09-000005 (v1), 9 October 2026. Click either reference number to open the registered submission.

[3] OpenAI, The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane Re s > 7/8 (30 September 2026). Official scope: lean/docs/003.md.

[Collet26] Hailey Collet, A zero-free half-plane with boundary 874861/1000000. https://haileystorm.github.io/research/cycle61-qrh-boundary/

[Liu26] Baiying Liu, Improved zero-free half-planes for the quasi-Riemann hypothesis, arXiv:2610.12234v2 (12 October 2026). https://arxiv.org/abs/2610.12234v2

Tao, Trudgian and Yang, arXiv:2501.16779, Table 2; Kerr, arXiv:1909.12075, Theorem 6, as identified in the checked source inventory. Exact source references and theorem identifiers are retained in BOUNDS.json and README_DRAFT.md.